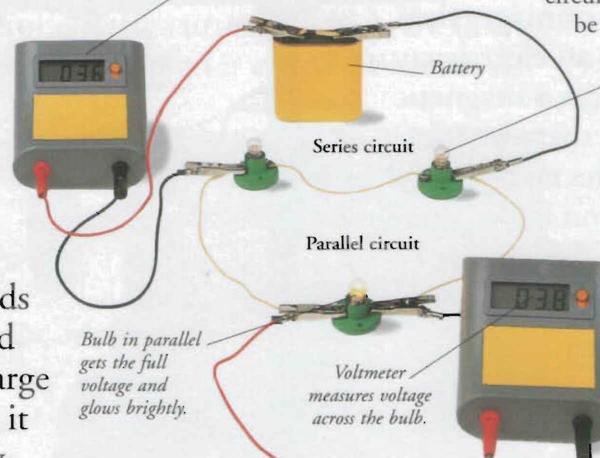
ELECTRICITY

A FLASH OF LIGHTNING is striking evidence of the invisible energy called electricity. This energy is produced by the movement of electrons – tiny particles found in atoms of matter. Every electron carries an identical negative electric "charge". When electric charge builds up in one place, it is called static electricity. If the charge flows from place to place, it is called current electricity.

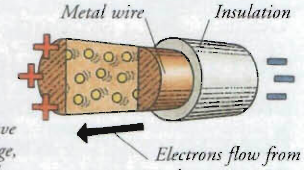
Electric circuit

The path around which current electricity flows is called a circuit. In the circuit shown here, electricity from the battery lights the bulbs. Two bulbs connected one after the other are described as being "in series". Bulbs in separate branches of the circuit are said to be "in parallel".



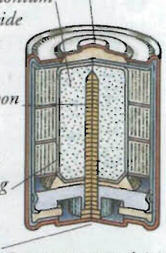
Electric current

Electrons pushed through the wires of a circuit form an electric current. The push on the electrons is called electromotive force (e.m.f). Voltage is a measure of e.m.f. The greater the voltage, the more current flows through the circuit.



Battery

A battery is a source of electric current. A chemical reaction between materials in the battery separates electrons from their atoms. The battery's e.m.f makes electrons flow out of the negative terminal, around a circuit, and back to the positive terminal.

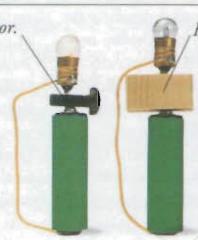


Static electricity

Rubbing two materials together can transfer electrons from one material to the other. A material that loses electrons gains a positive charge of static electricity, and a material that gains electrons gets a negative charge.

Attracting and repelling
A positively charged balloon attracts electrons to the surface of nearby hairs, giving them a negative charge. Opposite charges attract, so the hairs are pulled towards the balloon. Charges of the same type repel (push each other away).

Lightning
A tremendous charge of static electricity builds up inside a storm cloud. A flash of lightning occurs when this charge is suddenly released as a powerful electric current.

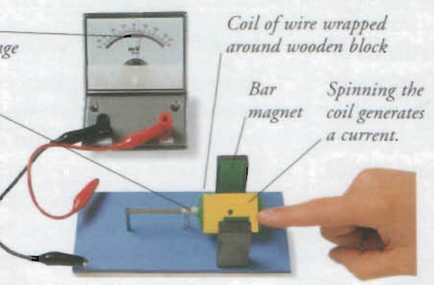


Conductors
Current can flow only through materials called conductors, whose electrons are bound loosely to their atoms and can be moved easily through the material.

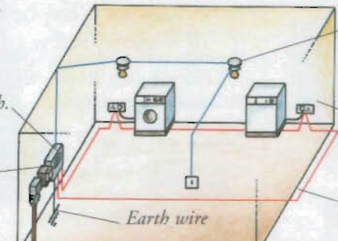
Insulators
Current cannot flow through insulators. The electrons in an insulator are bound firmly to their atoms and cannot move through the material.

Generator

Most of the electricity used in homes and factories is produced by devices called generators. Inside a generator, coils of wire spin rapidly in a magnetic field. The magnetism moves electrons through the wire, creating an electric current. In this simple version, bar magnets produce the magnetic field.



Simple electricity generator



Electricity supply
Electric current produced by generators in power stations reaches consumers via cables buried underground or carried by tall towers called pylons. The current alternates, which means that it changes direction many times each second. A battery produces direct current, which flows in one direction only.

Electricity in the home
Separate circuits in the home supply different voltages for different purposes. An electrical appliance takes power from the circuits through a plug that fits into a wall socket. The sockets are linked to the ground outside by an earth wire. If an electrical fault occurs, the current is diverted safely into the ground.

Michael Faraday

In 1831, the English scientist Michael Faraday (1791–1867) built the first generator after noticing that moving a magnet in and out of a wire coil made a current flow through the wire. Faraday also invented the electric motor and pioneered electrolysis (using electricity to break down substances).

Timeline

500s BC The ancient Greeks discover static electricity when they notice that amber (fossilized tree sap) attracts small objects if rubbed with wool.

Charged amber attracting feather

1752 American scientist and politician Benjamin Franklin proves that lightning is an electrical phenomenon.

1799 Italian physicist Alessandro Volta makes the first battery.

Volta's battery

1831 American physicist Joseph Henry and English Michael Faraday independently build "induction coils" – the first electricity generators.

1868 French chemist Georges Leclanché invents the Leclanché cell, the forerunner of modern zinc-carbon batteries.

1897 English physicist Joseph John Thomson discovers the electron.

FIND OUT MORE: ACIDS AND ALKALIS • ELECTROMAGNETISM • ENERGY • FRICTION • MAGNETISM • STORMS